

Regression Exercise Sol

April 2, 2026

0.1 Exercise

Using the California housing dataset, predict house prices based on:

- Average income
- Number of rooms
- Population
- Latitude and longitude

```
[6]: from sklearn.datasets import fetch_california_housing
data=fetch_california_housing()
X=data.data
Y=data.target
```

```
[ ]: # - MedInc      median income in block group
# - HouseAge     median house age in block group
# - AveRooms     average number of rooms per household
# - AveBedrms    average number of bedrooms per household
# - Population   block group population
# - AveOccup     average number of household members
# - Latitude     block group latitude
# - Longitude    block group longitude
```

```
[19]: # Split the data into training and testing
from sklearn.model_selection import train_test_split
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.2)
```

```
[34]: # Build the model
# from sklearn.linear_model import LinearRegression
# model = LinearRegression()

# from sklearn.linear_model import Ridge
# model = Ridge(alpha=1)

from sklearn.linear_model import Lasso
model = Lasso(alpha=1)

# from sklearn.linear_model import ElasticNet
model = ElasticNet(alpha=1, l1_ratio=.5)
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[35]: # Fit the model
model.fit(X_train, Y_train)
```

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[35]: Lasso(alpha=1)
```

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[36]: # Evaluate our model
y_predicted=model.predict(X_test)
from sklearn.metrics import mean_squared_error
mean_squared_error(Y_test, y_predicted)
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[36]: 0.9745013296699476
```

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[37]: from sklearn.metrics import r2_score
r2_score(Y_test, y_predicted)
```

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[37]: 0.28343433263500517
```

```
[39]: import numpy as np
import pandas as pd
new_house=pd.DataFrame(np.array([[8 ,42., 5.98,1.05, 322. ,2.56,37.88,-122.23],
                                [7 ,44., 7.7,2.0, 211. ,1.4, 10.78, -100.3]]),
    columns=data.feature_names)
new_house
```

```
[39]:   MedInc  HouseAge  AveRooms  AveBedrms  Population  AveOccup  Latitude  \
0     8.0     42.0     5.98      1.05      322.0      2.56      37.88
1     7.0     44.0     7.70      2.00      211.0      1.40      10.78

   Longitude
0    -122.23
1    -100.30
```

```
[40]: model.predict(new_house.values)
```

```
[40]: array([2.73423759, 2.60364036])
```

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[ ]:
```